

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 14



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Recap of Previous Lecture



Topic

Connectivity (Questions)



Topic

Euler trail , Euler Circuit and Traversable graph



Topic

Hamiltonian path, and Hamiltonian Circuit



Topics to be Covered



Topic

Distance, Eccentricity, Diameter, Radius and Girth



Topic

Spanning tree



Q:- Let K_n be a complete graph where $n \geq 3$
then how many Hamiltonian cycles exist in K_n .

↳ Every Hamiltonian cycle will contain
all 'n' vertices of complete graph K_n ,
∴ We are looking for different cycles
of length = n.

- (a) 1
- (b) $(n-1)!$
- (c) $\frac{(n-1)!}{2}$
- (d) $n!$

(i) → If vertices of graph are not labeled, then Ans = 1
because all cycles of length n are identical
if vertices are not labeled.

(ii) → If vertices of the graph are labeled, then Ans = $\frac{(n-1)!}{2}$

$n!$

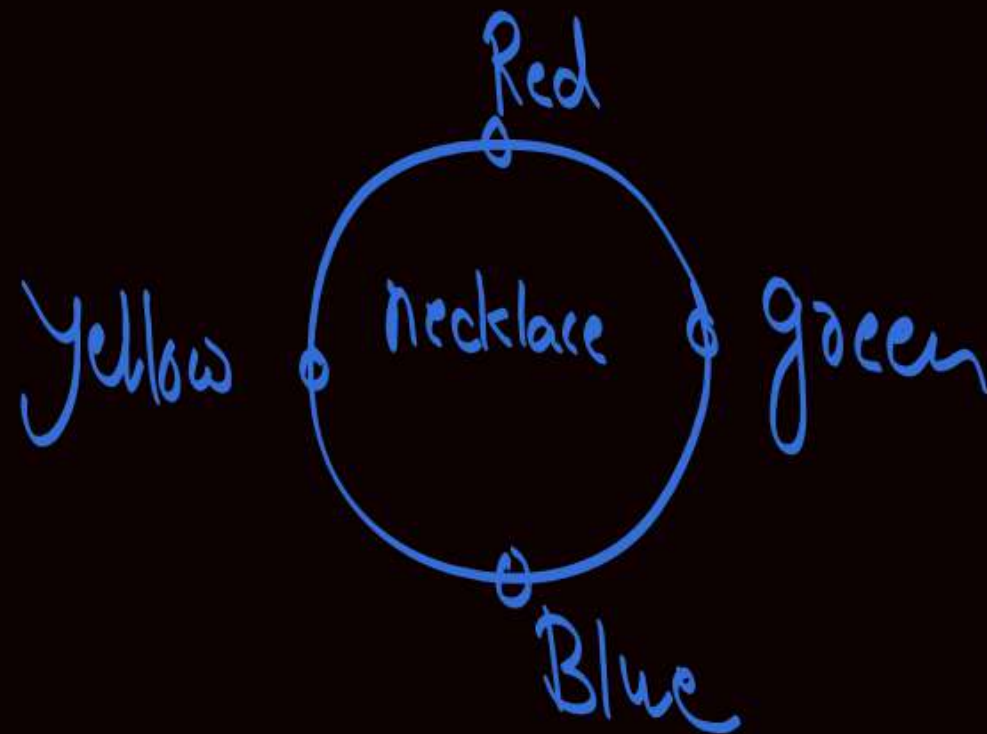
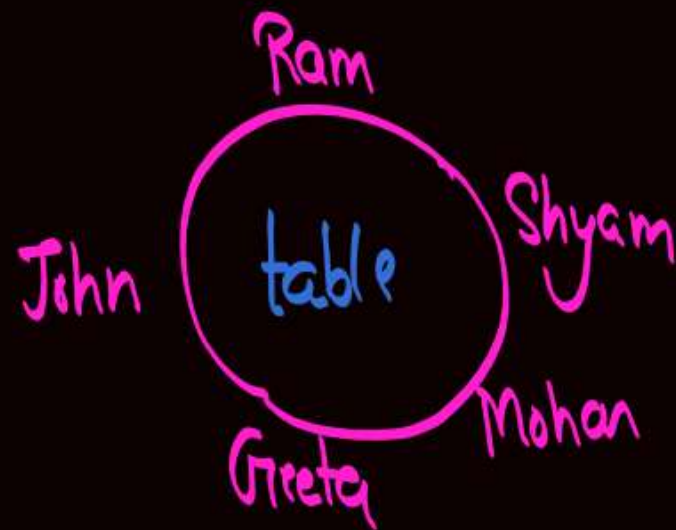
No. of linear permutations of 'n' distinct object

$(n-1)!$

No. of Circular permutations of 'n' objects

When Clockwise and Anticlockwise are different

{ eg. No. of Circular permutation of 'n' people }
around a Circular table



$\frac{(n-1)!}{2}$

= No. of Circular permutations of 'n' objects
When Clockwise and Anticlockwise are
Same



Topic : Distance

➤ The distance between two vertices of graph G is defined as number of edges on the shortest
Path between those two vertices.

distance
b/w

$$a-b = 3$$

$$a-c = 4$$

$$a-d = 6$$

$$a-e = 1$$

$$a-f = 2$$

$$a-g = 3$$

$$a-h = 4$$

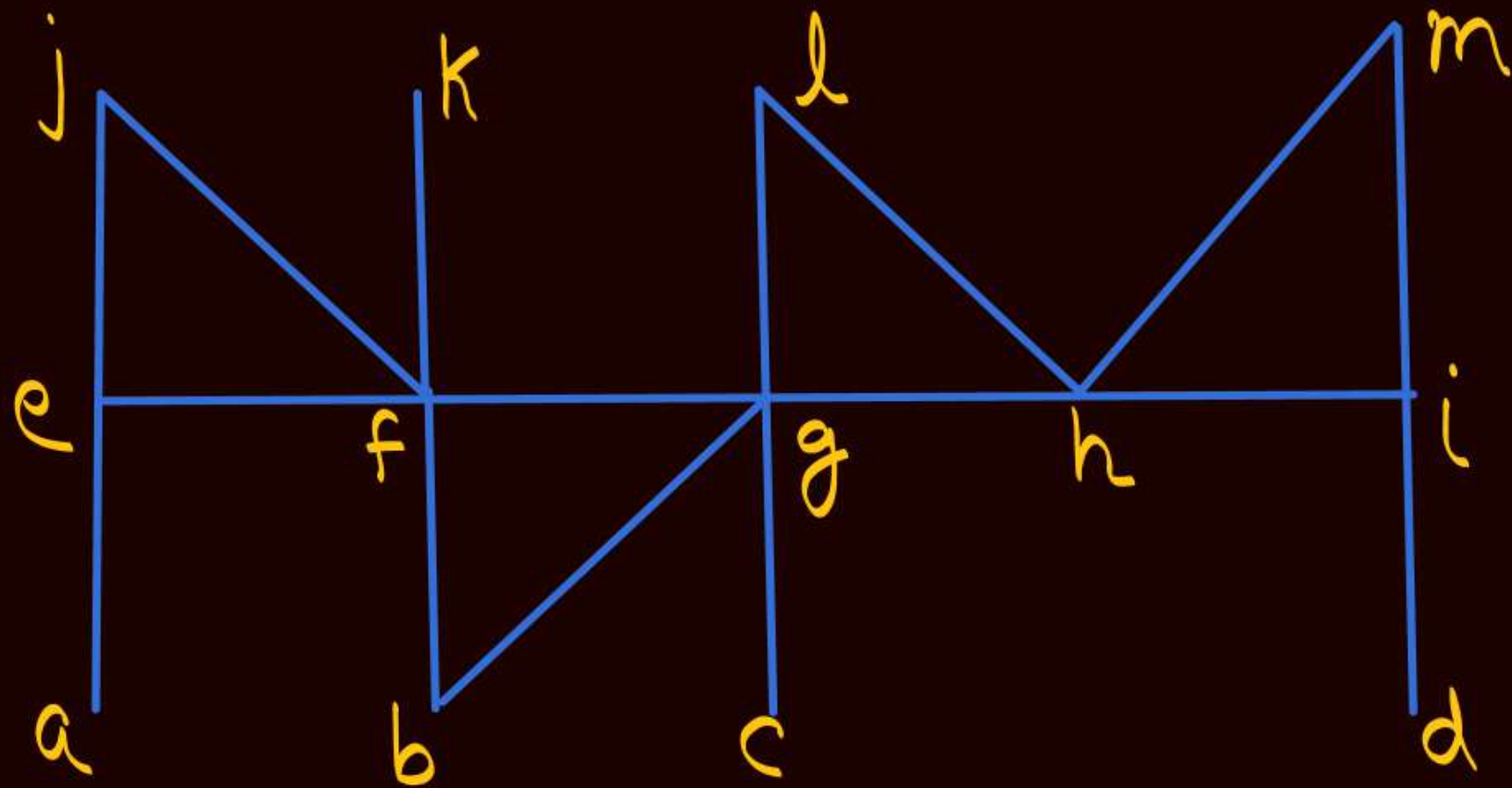
$$a-i = 5$$

$$a-j = 2$$

$$a-k = 3$$

$$a-l = 4$$

$$a-m = 5$$

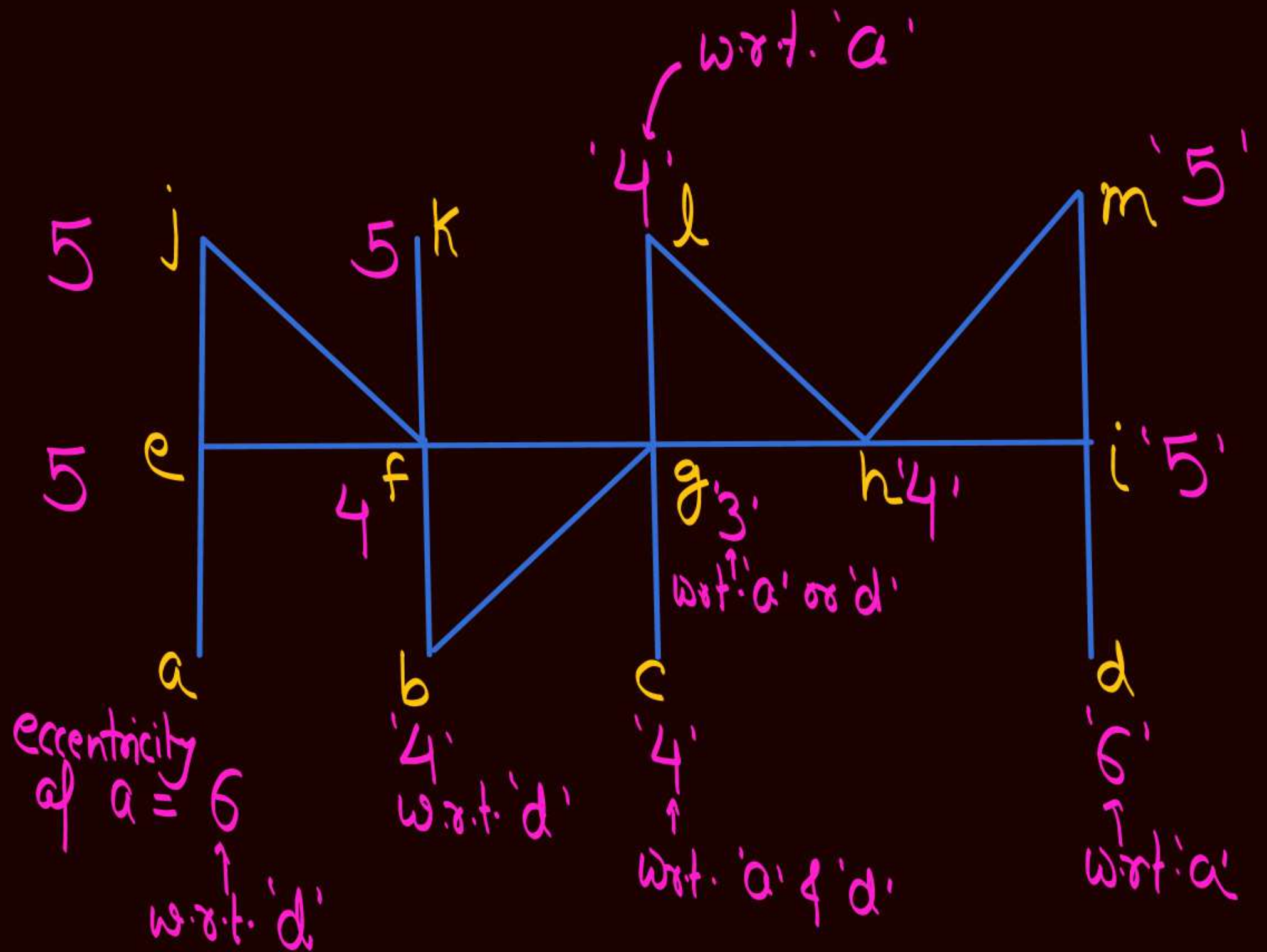




Topic : Eccentricity

Eccentricity of a vertex $v \in G$, is the maximum distance of vertex v from some other vertex of graph G .

Maximum Eccentricity = 6
Minimum Eccentricity = 3





Topic : Diameter



{ Diameter of graph G is defined as distance between farthest vertices of graph G }

→ Diameter of graph G is defined as maximum of the eccentricities of all the vertices of graph G .

→ In the above example,
diameter of graph $G = 6$



Topic : Radius



Radius of graph G is defined as minimum of the eccentricities of all the vertices of graph G .

- In the above example,
Radius of graph $G = 3$



Topic : Centre



→ The set of all vertices with minimum eccentricity is called Centre of the graph.

• In the above example,
Centre of graph:- $\{g\}$

→ There may be more than one vertex with minimum eccentricity, in that case Centre will be defined using a set containing all vertices with minimum eccentricity.



Topic : Girth



- In graph theory, the girth of an undirected graph is the length of a shortest cycle contained in the graph. If the graph does not contain any cycles (that is, it is a forest), its girth is defined to be infinity.

Spanning Tree



Topic : Spanning Tree



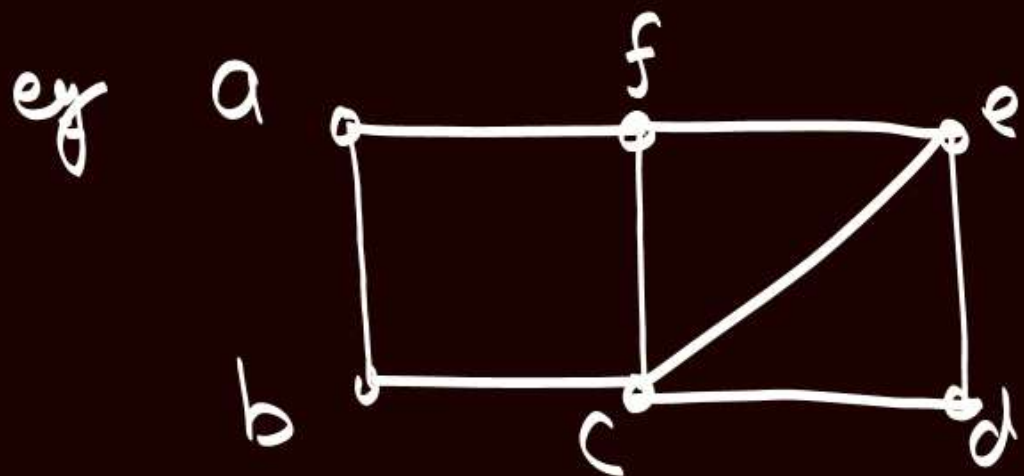
Let G be a connected graph,

If graph is not connected,
then spanning tree does not
exist for that graph

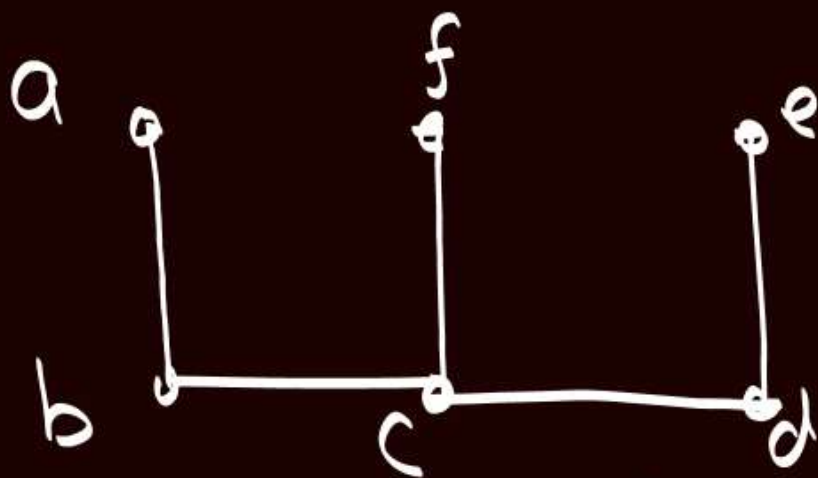
A sub-graph H of graph G is called a spanning tree if and only if

- H is a tree {i.e., acyclic + Connected}
- H contains all vertices of graph G

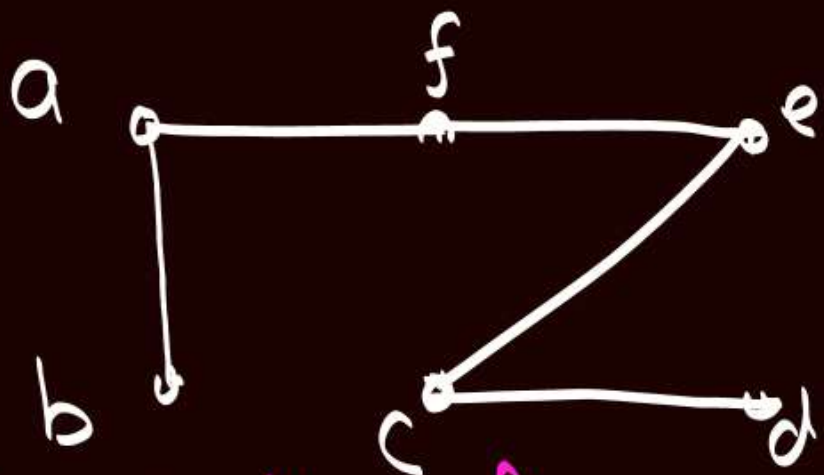
A graph G is a connected graph
if and only if
spanning tree exist for graph G



G



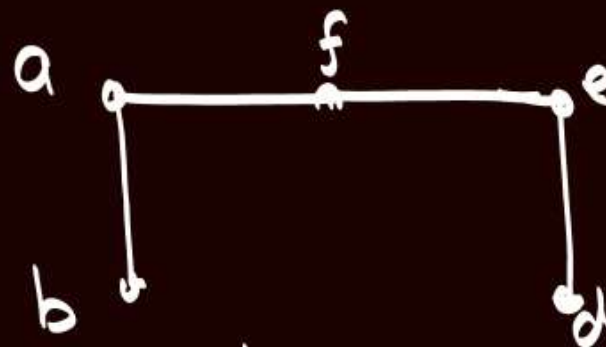
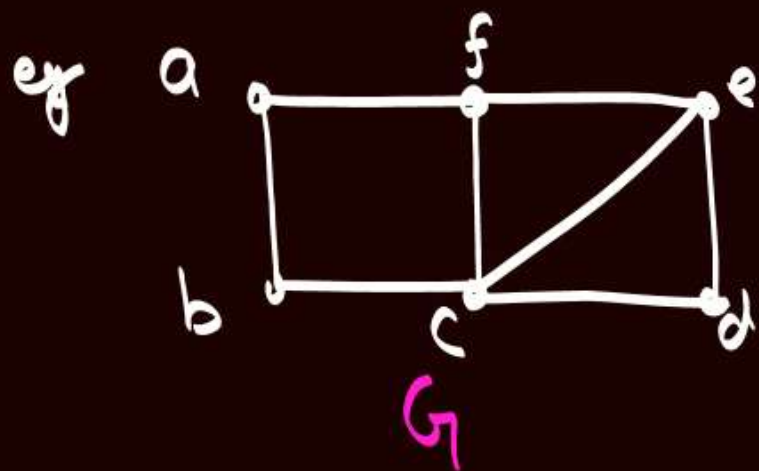
Sub-graph H_1 of graph G
 Acyclic ✓ + Connected ✓ + Contain all vertices ✓
 $\therefore H_1$ is a spanning tree of G



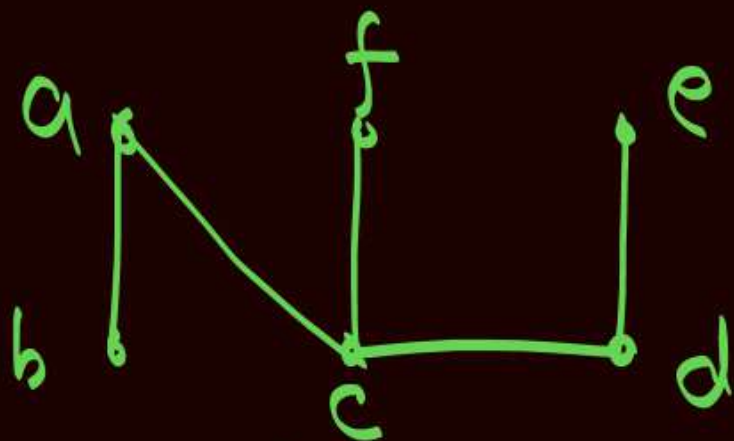
Sub-graph H_2 of graph G

Acyclic ✓ + Connected ✓ + Contain all vertices ✓
 $\therefore H_2$ is a spanning tree of G

For a connect graph G ,
 Spanning tree of G
 need not be unique



- ' H_3 '
- it is a subgraph of G ✓
 - it is a tree ✓
 - It is acyclic ✓
 - It is connected ✓
 - Contains all vertices of G ✗



- H_4
- it is a tree ✓
 - It is acyclic ✓
 - It is connected ✓
 - Contains all vertices of G ✓

→ It is not a sub graph of G . ∴ Not a spanning tree of G

∴ Not a spanning tree of G .

② If graph G is connected it will have at least '1' spanning tree

③ Graph G will have exactly '1' spanning tree if and only if graph G itself is a tree.

{ in that case, spanning tree will be graph G itself }

Note:

Let G be a connect graph with n -vertices,
then each spanning tree of G will contain
exactly $(n-1)$ edges.



Topic : Circuit Rank

Number of edges that must be deleted from a connected graph G in order to obtain a spanning tree of G is called circuit rank of graph G .

• Let G is a Connected graph with n -vertices and m -edges.

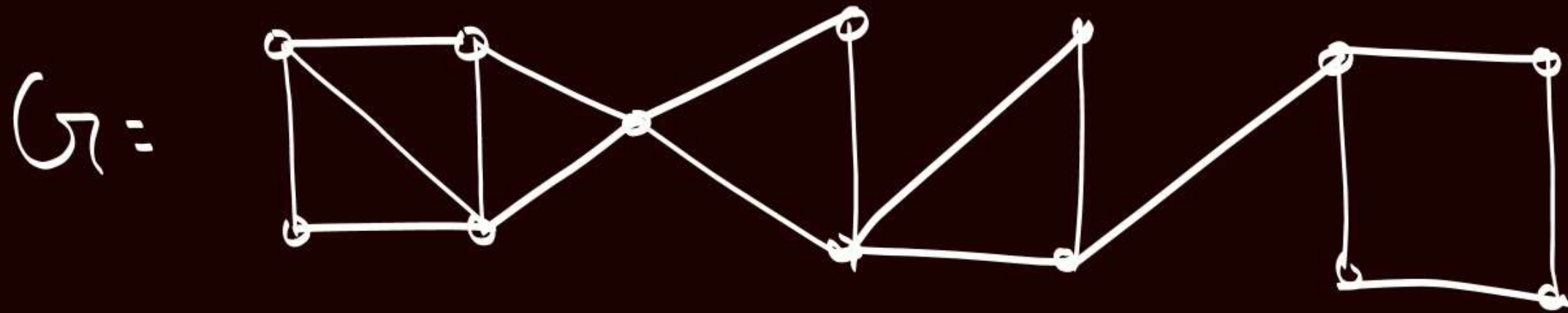
then

$$\text{Circuit Rank of graph } G = m - (n-1) = (m-n+1)$$

Total No. of edges
in G

No. of edges
Required in
Spanning tree

eg Find the Circuit rank of the following graph.

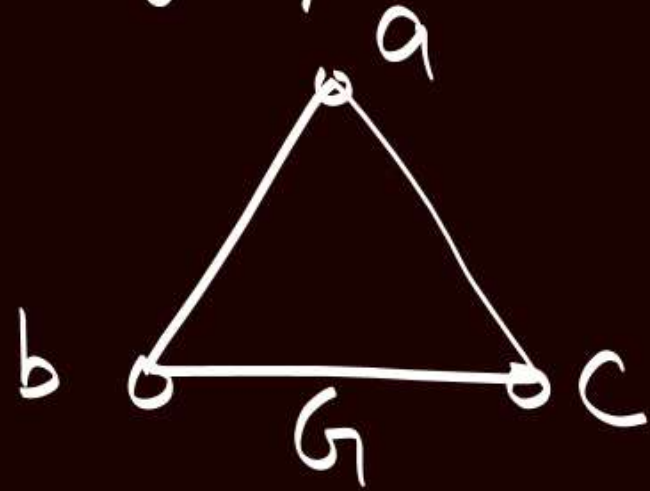


No. of vertices in $G = 13$

No. of Edges in $G = 18$

$\therefore$ Circuit rank of $G = 18 - (13 - 1)$
 $= 6$

Q: Find the number of spanning trees of the following graph

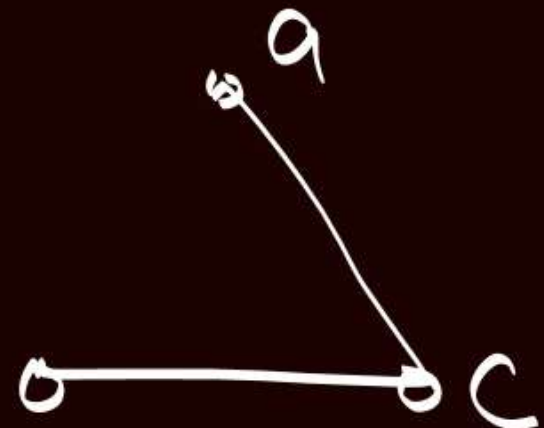
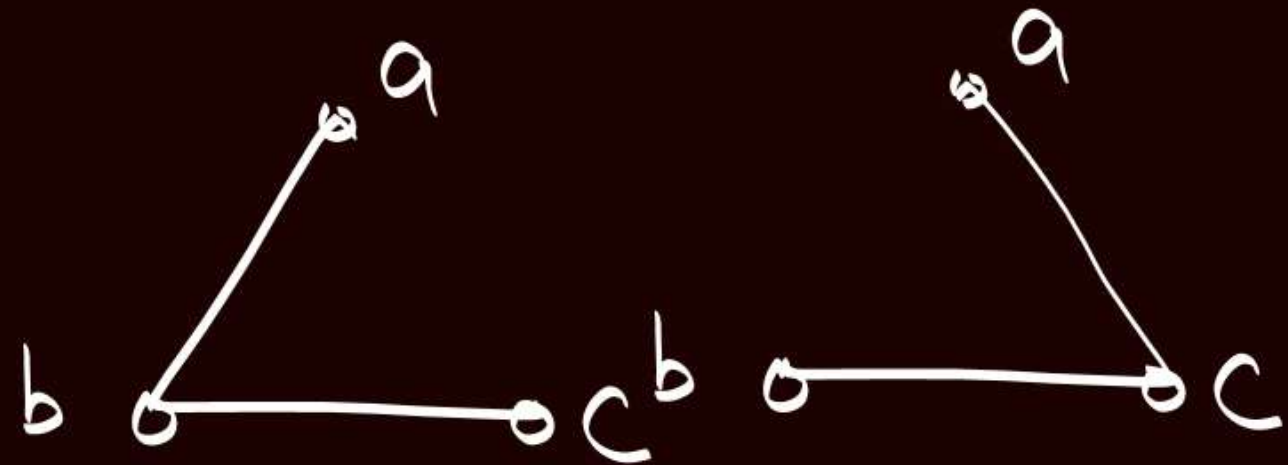
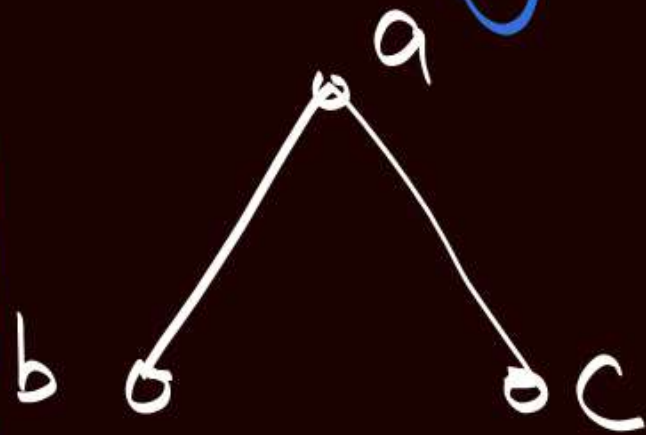


Circuit Rank of G : $3 - (3 - 1)$
 $= 1$

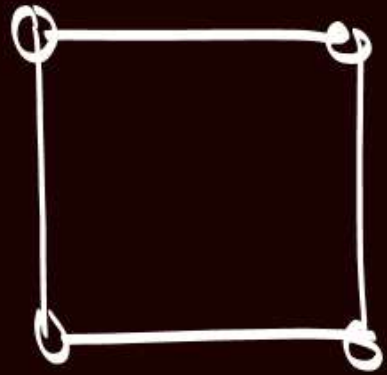
'1' edge from a cycle of length '3' can be deleted in '3' ways

$\therefore$ '3' spanning trees are possible

And they are



Q: Find the number of spanning trees of the following graph

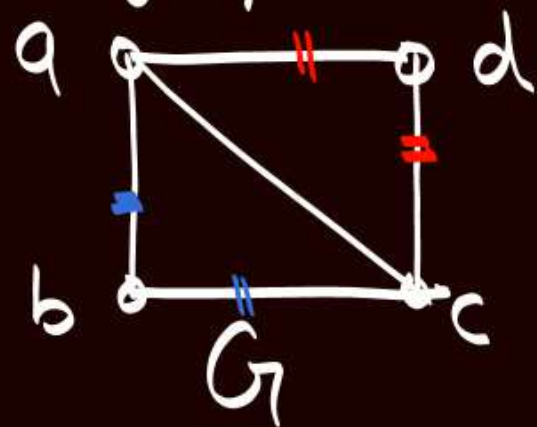


"G"

$$\text{Ans} = 4$$

$$\begin{aligned} \text{Circuit Rank of } G &= 4 - (4 - 1) \\ &= 1 \end{aligned}$$

Q: Find the number of spanning trees of the following graph



→ Out of '5' edges of graph 'G', two edges can be deleted in $5C_2 = 10$ ways

Each Combination of two edges need not produce a spanning tree

Circuit Rank of $G = 5 - (4 - 1) = 2$

Total no. of spanning tree

= Total no. of ways in which 2 out of 5 edges can be deleted

= $5C_2$

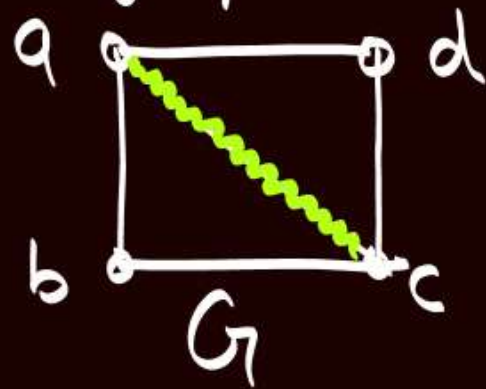
— No. of combinations of two edges such that deletion will not result in a spanning tree

— 2

One with $\{a,d\}$ & $\{c,d\}$

another with $\{a,b\}$ & $\{b,c\}$

Q: Find the number of spanning trees of the following graph



In a spanning tree of G

'either'
diagonal edge
will not be present

there is only '1' way to delete the diagonal edge
after that we will be left with a cycle of length = 4, $\therefore$ '4' ways to break the cycle

Spanning trees
such that diagonal edge is not present = $1 \times 4 = 4$

$\therefore$ Total no. of spanning trees = $4 + 4 = 8$

'or'
diagonal edge
will be present

there is only '1' way in which diagonal will be present (i.e., don't delete it)

In that case we need to delete '1' edge from $\{a,b\}, \{b,c\}$

(and) we need to delete '1' edge out of $\{a,d\}, \{c,d\}$

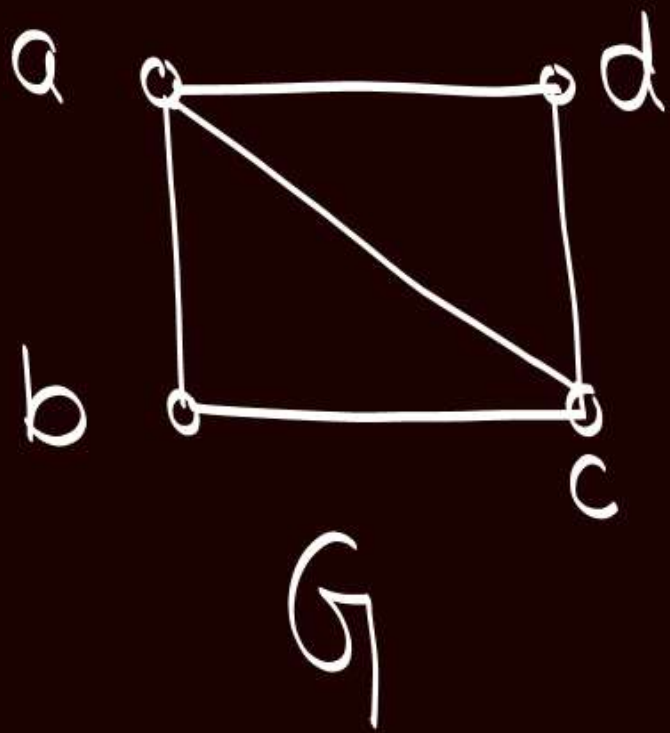
Spanning trees
such that diagonal edge is present = $1 * ({}^2C_1 * {}^2C_1) = 4$

* Kirchoff's Matrix Theorem:-

Kirchoff's matrix can be used to find the number of spanning trees of a given graph.

★ Kirchoff's Matrix Theorem:- Adjacency matrix is a way of representing the graph

→ Let 'A' is the adjacency matrix w.r.t. given graph



⇒ Adjacency matrix w.r.t. graph G

$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

* Kirchoff's Matrix Theorem:-

- 'M' is a matrix obtained from adjacency matrix 'A' by replacing all 1's of matrix A by '-1' and by replacing all the 0's of principle diagonal of matrix 'A' by the degrees of corresponding vertices.
} All other 0's of matrix 'A' will remain as it is }

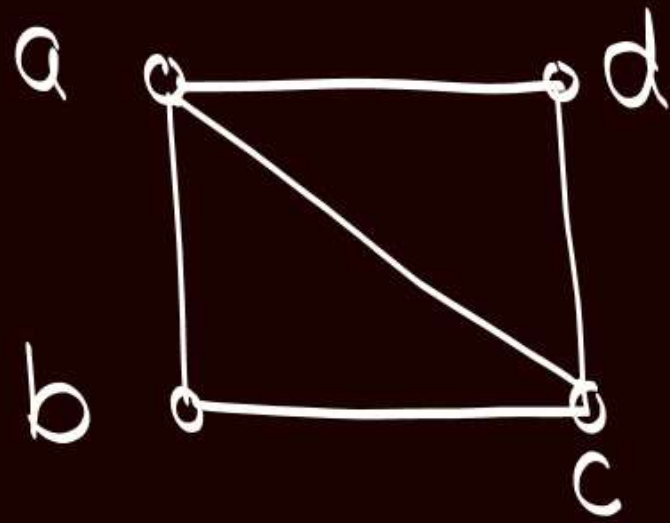
$$M = D - A$$

degree matrix
w.r.t. graph

Adjacency matrix
w.r.t. graph

} Matrix M is called }
Laplacian matrix }

★ Kirchoff's Matrix Theorem:-



G

$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$\Rightarrow$

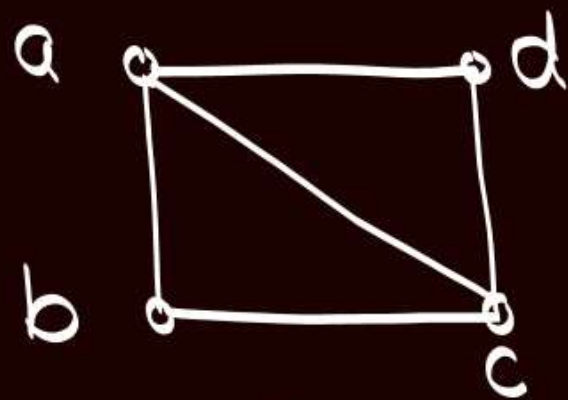
$$M = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \end{matrix}$$

deg(a) points to the value 3 in the first row, first column of M.

deg(b) points to the value 2 in the second row, second column of M.

★

Kirchoff's Matrix Theorem:-



⇒ Degree matrix
w.r.t. graph G:

$$D = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \end{matrix}$$

$$M = D - A$$

$$M = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \end{matrix} -$$

$$\begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix} =$$

$$\begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & 1 & 0 \\ -1 & 1 & 3 & 1 \\ -1 & 0 & 1 & 2 \end{bmatrix} \end{matrix}$$

• Laplacian matrix is a symmetric matrix

• In laplacian matrix, summation of the elements of each row and/or each column is always "Zero"

∴ Laplacian matrix is a singular matrix

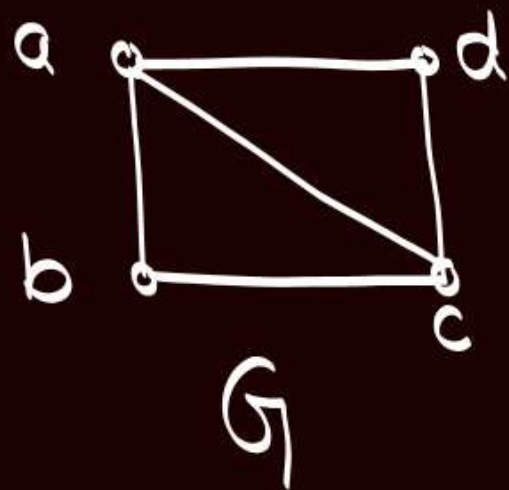
i.e. determinant value = 0

★ Kirchoff's Matrix Theorem:-

→ Once we have obtained matrix M ,
Co-factor w.r.t. any element of matrix M
will give us the number of spanning trees
Corresponding to the given graph

Co-factor of $a_{ij} = (-1)^{i+j} \times$ determinant value of the matrix
while skipping element of i^{th} row
and skipping the elements of j^{th} column.

★ Kirchhoff's Matrix Theorem: -



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$



$$M = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \end{matrix}$$

"a₁₁"

Co-factor of element "a₁₁" = $(-1)^{1+1} \star \begin{vmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix} = 1 \star \{ 2 \star (6-1) + 1(-2-0) + 0 \star (1-0) \}$

$= 1 \star \{ 10 + (-2) + 0 \}$

$= \textcircled{8} \text{ Ans}$

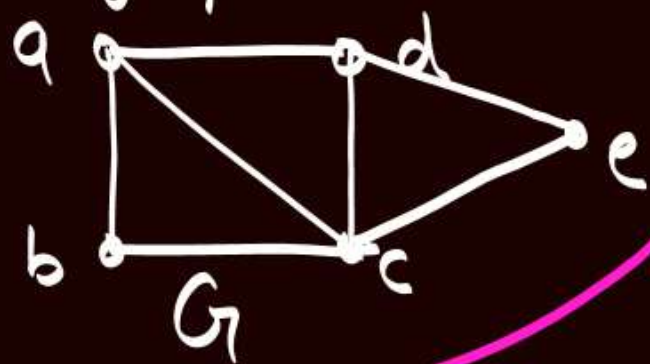
* Kirchoff's Matrix Theorem:-

Step 1: Write adjacency matrix 'A' w.r.t. given graph

Step 2: Obtain laplacian matrix 'M' from matrix 'A' by replacing 1's in matrix A by '-1' and by replacing 0's of principle diagonal in matrix 'A' by the degree of corresponding vertex

Step 3: Co-factor w.r.t. any element of matrix 'M' will give us the number of spanning tree

Q: Find the number of spanning trees of the following graph



$$M = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 4 & 1 & 1 \\ -1 & 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{bmatrix} \end{matrix}$$

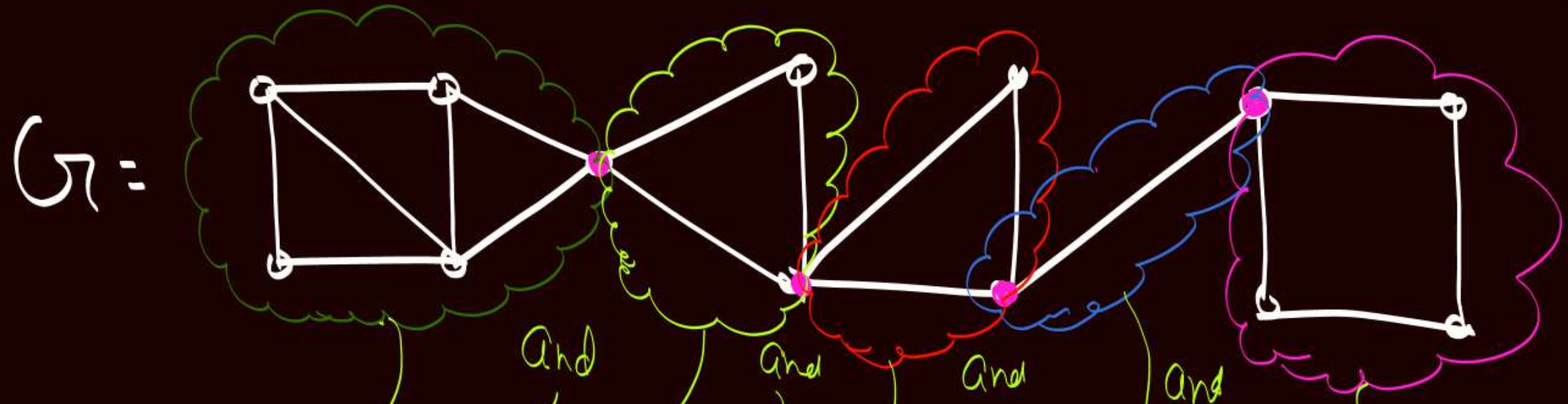
$$\begin{aligned} \text{Co-factor w.r.t. } a_{11} &= (-1)^{1+1} \times \begin{vmatrix} 2 & -1 & 0 & 0 \\ -1 & 4 & -1 & -1 \\ 0 & -1 & 3 & -1 \\ 0 & -1 & -1 & 2 \end{vmatrix} = 2 \times \begin{vmatrix} 4 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 2 \end{vmatrix} - (-1) \begin{vmatrix} -1 & -1 & -1 \\ 0 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix} \end{aligned}$$

$$= 2 \times [4(6-1) + 1(-2-1) - 1(1+3)] + 1[-1(6-1)]$$

$$= 2 \times (20 - 3 - 4) + 1 \times (-5)$$

$$= \underline{\underline{21}} \text{ Ans}$$

eg Find the number of spanning trees of the following graph



Total no. of
spanning tree
of graph G

$=$

21

*

3

*

3

*

1

*

4

$=$

756

Ans



Topic : Caley's Formula

→ For a Complete graph K_n
number of spanning trees can be given by $= n^{(n-2)}$

"Gate-2024" How many spanning trees are possible
w.r.t. a Complete graph with '4' vertices

$$\Rightarrow \underline{\underline{Ans}} = (4)^{(4-2)} = (4)^2 = \underline{\underline{16}}$$



2 mins Summary



Topic

Distance, Eccentricity, Diameter, Radius and Girth



Topic

Spanning tree

THANK - YOU